

PCAP Analysis Write-Up

Name: ABDUL SABOOR

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Exercise Name: Nemotodes

Source: malware-traffic-analysis.net

Infected IP

Statistics → Conversations → IPv4 tab → sort by Packets

Ethernet · 6	IPv4 · 60	IPv6	TCP · 142	UDP · 134							
Address A	Address B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B
10.11.26.183	193.42.38.139	13,248	11 MB	21	5,648	358 kB	7,600	11 MB	35.862131	31.3138	91 kbps

Infected IP=10.11.26.183

MAC Address of Infected IP:

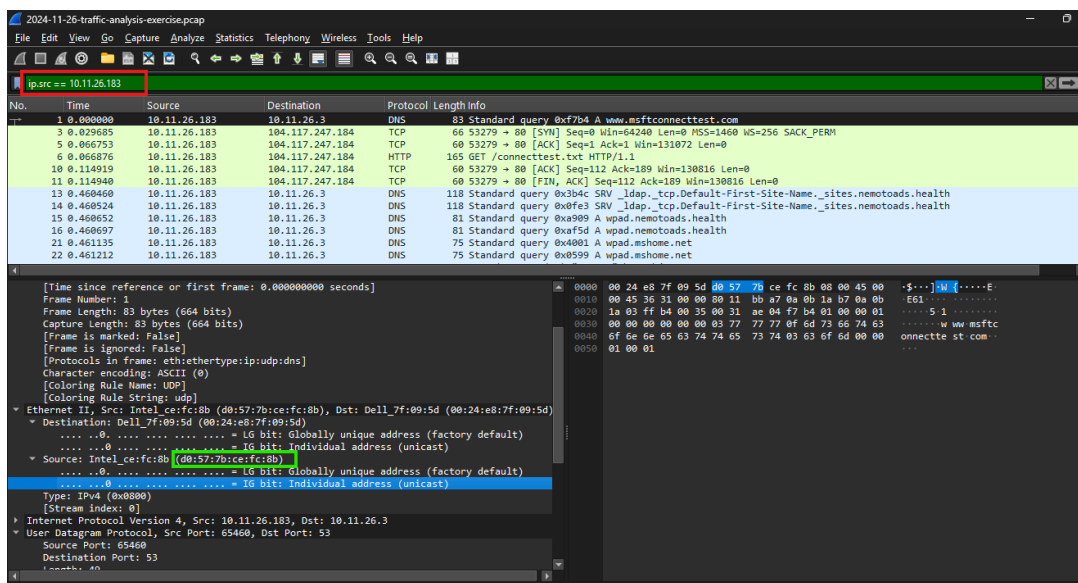
Method:

Step 1: Filter ip.src == 10.11.26.183

Step 2: Click any packet

Step 3: Middle panel → expand Ethernet II

Step 4: Look at Src field



Result: d0:57:7b:ce:fc:8b

Hostname

Method:

Step 1: Try filter bootp first , if nothing appears

Step 2: Try filter nbns

Step 3: Click any packet from 10.11.26.183

Step 4: Goto Middle panel then look at Name field in NBNS section

The image shows a Wireshark packet capture interface. The top panel displays a list of packets filtered by 'nbns'. The middle panel shows the details of a selected packet (No. 145) in the NetBIOS Name Service section. The bottom panel shows the packet bytes in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
69	2.866033	10.11.26.183	10.11.26.255	NBNS	110	Registration NB DESKTOP-B8TQK49<20>
70	2.866132	10.11.26.183	10.11.26.255	NBNS	110	Registration NB DESKTOP-B8TQK49<00>
71	2.866203	10.11.26.183	10.11.26.255	NBNS	110	Registration NB NEMOTODES<00>
144	3.628777	10.11.26.183	10.11.26.255	NBNS	110	Registration NB NEMOTODES<00>
145	3.628873	10.11.26.183	10.11.26.255	NBNS	110	Registration NB DESKTOP-B8TQK49<00>
146	3.628873	10.11.26.183	10.11.26.255	NBNS	110	Registration NB DESKTOP-B8TQK49<20>
174	4.405818	10.11.26.183	10.11.26.255	NBNS	110	Registration NB DESKTOP-B8TQK49<20>
175	4.405818	10.11.26.183	10.11.26.255	NBNS	110	Registration NB DESKTOP-B8TQK49<00>
176	4.405945	10.11.26.183	10.11.26.255	NBNS	110	Registration NB NEMOTODES<00>
180	5.161747	10.11.26.183	10.11.26.255	NBNS	110	Registration NB NEMOTODES<00>
181	5.161747	10.11.26.183	10.11.26.255	NBNS	110	Registration NB DESKTOP-B8TQK49<00>
182	5.161878	10.11.26.183	10.11.26.255	NBNS	110	Registration NB DESKTOP-B8TQK49<20>

Character encoding: ASCII (0)
[Coloring Rule Name: SMB]
[Coloring Rule String: smb || nbss || nbns || netbios]

- ▶ Ethernet II, Src: Intel_ce:fc:8b (d0:57:7b:ce:fc:8b), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
- ▶ Internet Protocol Version 4, Src: 10.11.26.183, Dst: 10.11.26.255
- ▶ User Datagram Protocol, Src Port: 137, Dst Port: 137
- ▼ NetBIOS Name Service
 - Transaction ID: 0xa116
 - ▶ Flags: 0x2910, Opcode: Registration, Recursion desired, Broadcast
 - Questions: 1
 - Answer RRs: 0
 - Authority RRs: 0
 - Additional RRs: 1
 - ▶ Queries
 - ▼ Additional records
 - ▼ DESKTOP-B8TQK49<00>: type_NB, class_IN
 - Name: DESKTOP-B8TQK49<00> (Workstation/Redirector)
 - Type: NB (32)
 - Class: IN (1)
 - Time to live: 3 days, 11 hours, 20 minutes
 - Data length: 6
 - ▶ Name flags: 0x4000, ONT: Unknown (M-node, unique)
 - Addr: 10.11.26.183

0000 ff ff ff ff ff ff d0 57
0010 00 60 32 c5 00 00 80 11
0020 1a ff 00 89 00 89 00 4c
0030 00 00 00 00 00 01 20 45
0040 45 45 50 46 41 43 4e 45
0050 4c 44 45 44 4a 41 41 00
0060 00 01 00 04 93 e0 00 06

Result: DESKTOP-B8TQK49

Username

Method:

Step 1: Filter kerberos

Step 2: Click any AS-REQ packet

Step 3: Middle panel → expand Kerberos → as-req → req-body → cname

Step 4: Read CNameString

The image shows a Wireshark network traffic capture. The top panel, 'Packet List', is filtered for 'kerberos'. It shows several packets, with frame 213 (time 16.964985) being an AS-REQ packet from 10.11.26.183 to 10.11.26.3. The middle panel, 'Packet Details', is expanded for frame 213. It shows the 'Kerberos' section, which is further expanded to show 'as-req'. Under 'as-req', the 'req-body' section is expanded, and the 'cname' field is selected. The 'cname' field contains 'name-type: kRB5-NT-PRINCIPAL (1)' and 'cname-string: 1 item'. The 'cname-string' field is expanded, showing 'CNameString: oboomwald'.

No.	Time	Source	Destination	Protocol	Length	Info
43	0.573783	10.11.26.183	10.11.26.3	LDAP	715	bindRequest(14) "<RC
45	0.576067	10.11.26.3	10.11.26.183	LDAP	265	bindResponse(14) suc
213	16.964985	10.11.26.183	10.11.26.3	KRB5	281	AS-REQ
214	16.965965	10.11.26.3	10.11.26.183	KRB5	245	KRB Error: KRB5KDC_E
221	16.970929	10.11.26.183	10.11.26.3	KRB5	361	AS-REQ
223	16.972980	10.11.26.3	10.11.26.183	KRB5	340	AS-REP
232	16.973914	10.11.26.183	10.11.26.3	KRB5	254	TGS-REQ
235	16.976443	10.11.26.3	10.11.26.183	KRB5	275	TGS-REP
268	17.125036	10.11.26.183	10.11.26.3	KRB5	406	TGS-REQ
271	17.125200	10.11.26.183	10.11.26.3	KRB5	424	TGS-REQ
275	17.126586	10.11.26.3	10.11.26.183	KRB5	395	TGS-REP
279	17.126764	10.11.26.3	10.11.26.183	KRB5	375	TGS-REP

Frame 213: Packet, 281 bytes on wire (2248 bits), 281 bytes captured (2248 bits)
Ethernet II, Src: Intel_ce:fc:8b (d0:57:7b:ce:fc:8b), Dst: Dell_7f:09:5d (00:24:e8:7f:09:5d)
Internet Protocol Version 4, Src: 10.11.26.183, Dst: 10.11.26.3
Transmission Control Protocol, Src Port: 53286, Dst Port: 88, Seq: 1, Ack: 1, Len: 227
Kerberos
 Record Mark: 223 bytes
 as-req
 pvno: 5
 msg-type: krb-as-req (10)
 padata: 1 item
 req-body
 Padding: 0
 kdc-options: 40810010
 cname
 name-type: kRB5-NT-PRINCIPAL (1)
 cname-string: 1 item
 CNameString: oboomwald
 realm: NEMOTODES
 sname

Result: oboomwald

Full Name

Step 1: Edit → Find Packet

Step 2: Search type: String, Field: Packet Bytes

Step 3: Type boomwald

Step 4: Find packet containing CN=Oliver Q. Boomwald

The image shows the Wireshark network protocol analyzer interface. At the top, a display filter 'boomwald' is applied to the packet list. The packet list shows several packets, with packet 21266 selected. The packet details pane shows the structure of the selected packet, including Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, and Lightweight Directory Access Protocol. The LDAP message details show a searchResEntry(4) for 'CN=Oliver Q. Boomwald, CN=Users, DC=nemotodes, DC=health' and a searchResDone(4) success [2 results].

No.	Time	Source	Destination	Protocol	Length	Info
21261	318.577955	10.11.26.183	10.11.26.3	TCP	1514	53395 → 389 [ACK] Seq=...
21262	318.578002	10.11.26.183	10.11.26.3	LDAP	604	bindRequest(3) "<ROOT>"
21263	318.578106	10.11.26.3	10.11.26.183	TCP	60	389 → 53395 [ACK] Seq=2...
21264	318.580286	10.11.26.3	10.11.26.183	LDAP	265	bindResponse(3) success
21265	318.581635	10.11.26.183	10.11.26.3	LDAP	218	SASL GSS-API Integrity:
21266	318.582512	10.11.26.3	10.11.26.183	LDAP	242	SASL GSS-API Integrity:
21267	318.582800	10.11.26.183	10.11.26.3	LDAP	97	SASL GSS-API Integrity:
21268	318.582995	10.11.26.3	10.11.26.183	TCP	60	389 → 53395 [RST, ACK]
21269	318.583103	10.11.26.183	10.11.26.3	TCP	60	53395 → 389 [FIN, ACK]
21270	318.583163	10.11.26.3	10.11.26.183	TCP	60	389 → 53395 [RST] Seq=3...
21271	318.588908	10.11.26.183	10.11.26.3	TCP	66	53396 → 389 [SYN] Seq=6...

Frame 21266: Packet, 242 bytes on wire (1936 bits), 242 bytes captured (1936 bits)

- Ethernet II, Src: Dell_7f:09:5d (00:24:e8:7f:09:5d), Dst: Intel_ce:fc:8b (d0:57:7b:ce:fc:8b)
- Internet Protocol Version 4, Src: 10.11.26.3, Dst: 10.11.26.183
- Transmission Control Protocol, Src Port: 389, Dst Port: 53395, Seq: 3011, Ack: 2525, Len: 188
- Lightweight Directory Access Protocol
 - SASL Buffer Length: 184
 - SASL Buffer
 - GSS-API Generic Security Service Application Program Interface
 - GSS-API payload (156 bytes)
 - LDAPMessage searchResEntry(4) "CN=Oliver Q. Boomwald, CN=Users, DC=nemotodes, DC=health"
 - LDAPMessage searchResDone(4) success [2 results]

Result: Oliver Q. Boomwald

Indicators of Compromise (IOCs)

Method:

Step 1: Filter http.request and ip.src == 10.11.26.183

Step 2: Look at Info column and analyze the packet

Filter: http.request and ip.src == 10.11.26.183

No.	Time	Source	Destination	Protocol	Length	Info
6	0.066876	10.11.26.183	104.117.247.184	HTTP	165	GET /connecttest.txt HTTP/1.1
82	3.006408	10.11.26.183	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
161	3.941531	10.11.26.183	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
192	6.012703	10.11.26.183	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
194	6.956644	10.11.26.183	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
198	7.935707	10.11.26.183	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
202	10.938503	10.11.26.183	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
205	13.938519	10.11.26.183	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
12788	62.375093	10.11.26.183	104.117.247.99	HTTP	294	GET /MFMwUTBPME0wSzA3BgUrDgMCGGUABBRpD%2BQVZ%2B1 HTTP/1.1
20336	67.286490	10.11.26.183	104.26.1.231	HTTP	172	GET /location/locas.asp HTTP/1.1
20340	67.391300	10.11.26.183	194.180.191.64	HTTP	274	POST http://194.180.191.64/fakeurl.htm HTTP/1.1
20342	67.588001	10.11.26.183	194.180.191.64	HTTP	502	POST http://194.180.191.64/fakeurl.htm HTTP/1.1

Packet 20336 details:

- Total Length: 158
- Identification: 0x22b9 (8889)
- 010. = Flags: 0x2, Don't fragment
- ...0 0000 0000 0000 = Fragment Offset: 0
- Time to Live: 128
- Protocol: TCP (6)
- Header Checksum: 0x48de [validation disabled]
- [Header checksum status: Unverified]
- Source Address: 10.11.26.183
- Destination Address: 104.26.1.231
- [Stream index: 37]
- Transmission Control Protocol, Src Port: 53363, Dst Port: 80, Seq: 1, Ack: 1, Len: 118
- Hypertext Transfer Protocol
 - GET /location/locas.asp HTTP/1.1\r\n
 - Request Method: GET
 - Request URI: /location/locas.asp
 - Request Version: HTTP/1.1
 - Host: geo.netsupportsoftware.com\r\n
 - Connection: keep-alive\r\n
 - Cache-Control: no-cache\r\n
 - \r\n

Answer:

IP: 194.180.191.64

URL: http://194.180.191.64/fakeurl.htm

Domain: geo.netsupportsoftware.com

Executive Summary:

On 2024-11-26, a Windows workstation at IP 10.11.26.183 within the nemotodes.health network was infected with NetSupport Manager RAT malware. The infected machine made repeated HTTP connections to the malicious IP 194.180.191.64 using the URL /fakeurl.htm

which is a known NetSupport RAT C2 signature. The machine also contacted geo.netsupportsoftware.com for victim geolocation

Victim Details:

Field	Value
IP Address	10.11.26.183
MAC Address	d0:57:7b:ce:fc:8b
Hostname	DESKTOP-B8TQK49
Username	oboomwald
Full Name	Oliver Q. Boomwald
Domain	nemotodes.health

Indicators of Compromise:

Type	Value
Malicious IP	194.180.191.64
Malicious URL	http://194.180.191.64/fakeurl.htm
RAT Domain	geo.netsupportsoftware.com
Malware	NetSupport Manager RAT

Note:

All files used are publicly available PCAP files for SOC training purposes only.

No malware was executed during this analysis.